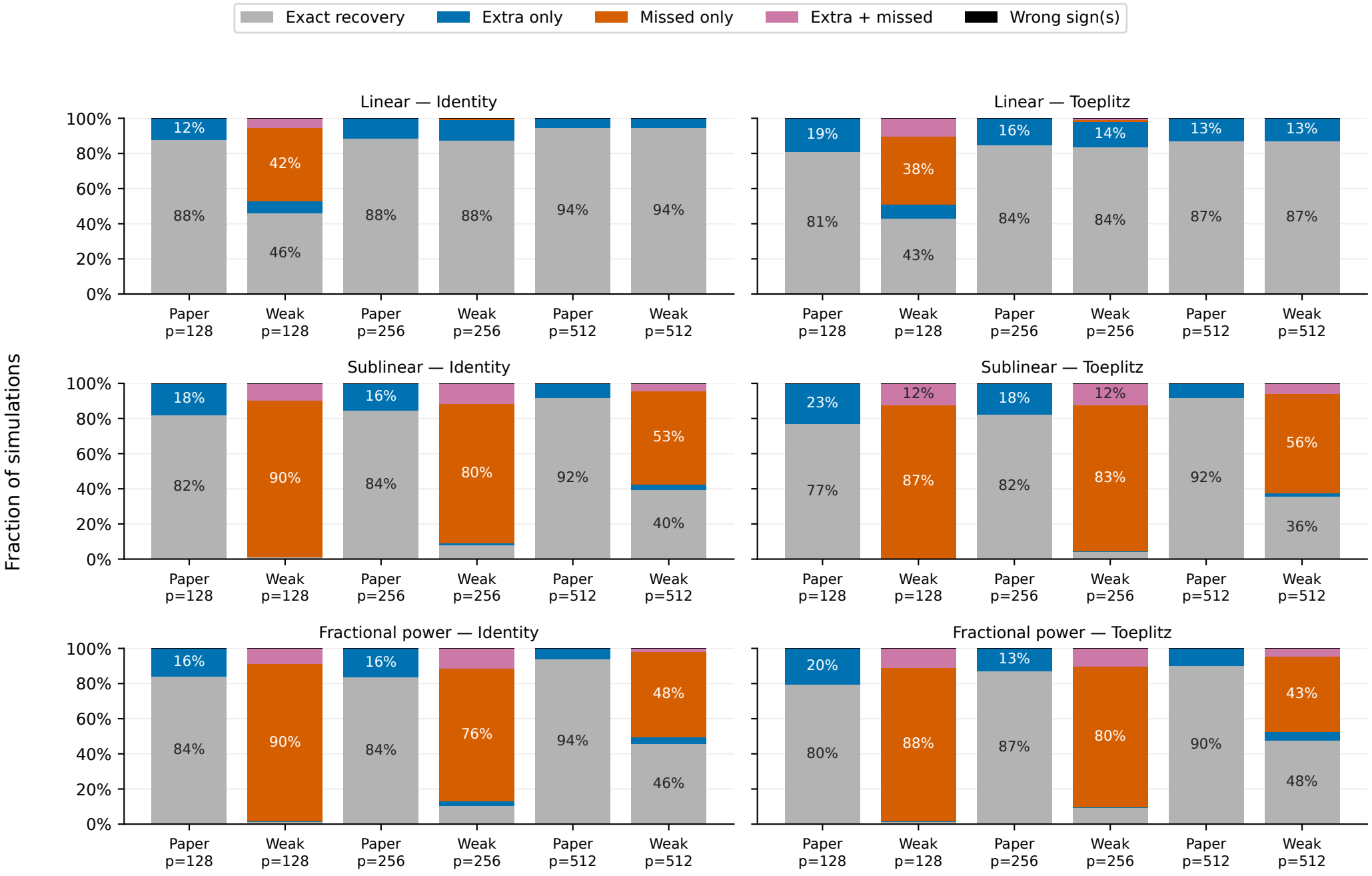
# Why exact Lasso recovery fails

$\theta = 1.3$; paper: active effects ±1; weak: five at ±0.25, rest ±1; noise SD $= 0.5$




200 simulation trials per bar. Exact recovery requires every support location and sign to be correct.
Extra = a selected null variable; missed = an omitted true variable. Wrong-sign cases take priority; all categories sum to 100%.
Gaussian predictors: independent (left) or Toeplitz with adjacent correlation 0.1 (right); random support and signs.
n = floor[2θk log(p − k)] + k + 1; sparsity uses α = 0.4, γ = 0.75. Frequencies are descriptive; segments ≥12% are labeled.